Romain Noblet - 05/06/2026

Conversational Clustering

Can natural-language feedback steer clustering toward the user's intent — even when the embedding's default similarity disagrees?

Most don't know what they want until they see what they don't.

Research Questions

HEADLINE

Does multi-turn natural-language feedback recover hidden target clusterings better than one-shot prompting — and does the advantage depend on whether the target conflicts with the embedding's defaults?

RQ1

Efficiency

Fewer turns to reach a target ARI threshold?

RQ2

Ceiling

Higher final ARI against the hidden target?

RQ3

Bias override

Does the gap grow when the target conflicts with embedding defaults?

RQ4

Stability

Stable across runs and corpus resamples?

Dataset · arXiv astro-ph

WHAT IT IS

arXiv astro-ph — the astrophysics preprint section of arXiv.org. Open-access papers with title, abstract, authors, and submitter-assigned category tags. (Astrophysics of Galaxies; Earth and Planetary Astrophysics, Solar and Stellar Astrophysics; …)

WORKING SUBSET

200
abstracts

6
primary categories

0.32
ARI(k-means vs tags)

Easy + hard targets

arXiv tags give a free topical target. Methodology / scale are hand-labelable in hours.

Evaluation metric · ARI

Why Adjusted Rand Index, and how to read the numbers

WHAT IT MEASURES

Adjusted Rand Index compares two clusterings of the same objects — here, the system's clustering and a hidden target labeling.

How: counts pairs of papers that the two clusterings agree about (both put them together OR both put them apart), corrected for what random chance alone would give.

READING THE NUMBERS

1.0	perfect agreement
0.5+	strong recovery of the target
0.3	partial recovery
0.0	no better than random
< 0	worse than random (rare)

Chance-corrected

Random clusterings get ~0, regardless of K. Lets us compare runs with different cluster counts.

Label-blind

ARI ignores cluster names. The system can label its 'galaxies' cluster anything; only membership matters.

Preliminary results · LLM baselines

Claude Sonnet 4.6 on 200 abstracts · the effect we're going to study, measured

ARI vs arXiv primary category

baseline	ARI	vs default
generic (no axis)	0.245	*free-form*
topic	0.361	*aligned*
object scale	0.264	*partial*
methodology	**0.071**	*orthogonal*

Spread = ***0.29*** *(topic − methodology)*

REFERENCE POINTS

How does Claude topic baseline (0.36) compare?

0.32	k-means (Week 1)
~0.03	llama3.2:3b (local)
0.36	**Claude Sonnet 4.6**

Claude follows axis instructions

topic (0.36) > generic (0.25) > methodology (0.07). The axis matters.

Methodology is truly orthogonal

methodology ↔ topic ARI = 0.05. Conflicts with embedding defaults.

Local model insufficient

llama3.2:3b ARI ≈ 0.03 even on topic. Claude-class capacity required.

Where this sits in the literature

LLM-guided clustering is crowded — the measurement angle isn't

Closest prior art

- **ClusterLLM (EMNLP 2023)**

  User 'perspective' + LLM-judged triplets refine embeddings.

- **ITGC (2025)**

  Iterative text-guided refinement — on images.

- **InBedder (2024)**

  Instruction-following embeddings: change what's encoded.

- **Dial-In LLM (2024)**

  LLM-in-the-loop intent clustering with refinement.

- **Viswanathan et al. (2024)**

  Few-shot LLM-guided clustering at three pipeline stages.

THE GAP I'M AIMING AT

Prior work mostly evaluates against targets that align with embedding defaults.

Week-1 finding: sentence-transformer embeddings encode multiple axes — topic, methodology, object scale. A user goal aligns with one and conflicts with others.

Week-2 baselines confirm the effect empirically: Claude one-shot reaches ARI = 0.36 on topic vs 0.07 on methodology — a 0.29 spread to close.

System architecture is not novel. The contribution is empirical, not architectural.

Initial experiment design

Simulated-user methodology · three hidden targets of varying difficulty

1
Experimenter
defines hidden target

2
Simulated user
gives natural language feedback toward target

3
System
feedback → ops on pipeline

4
New clustering
ops applied, clustering updated

5
ARI logged
output vs hidden target

↻ *Loop 6 times*

Three hidden targets · varying difficulty

EASY

Topic

arXiv primary categories.

Claude one-shot reaches 0.36

HARD

Object scale

Planetary / stellar / galactic / cosmological.

Claude one-shot reaches 0.26

HARDEST

Methodology

Observational / theoretical / simulation / instrumental.

Claude one-shot reaches 0.07

System architecture

LLM steers classical clustering through a defined operation vocabulary

arXiv astro-ph corpus

~200 abstracts

Sentence-transformer

all-MiniLM-L6-v2

k-means clustering

current K, current assignments

↑ ***LLM summary sent back to user***

Cluster summarizer (LLM)

short per-cluster description for the user

→

LLM router

OPERATION VOCABULARY

Free-form feedback ⟶ typed operations on the pipeline.

`change_k(k)`
re-run k-means with new K

`split(c)`
sub-cluster a single cluster

`merge(a, b)`
combine two clusters

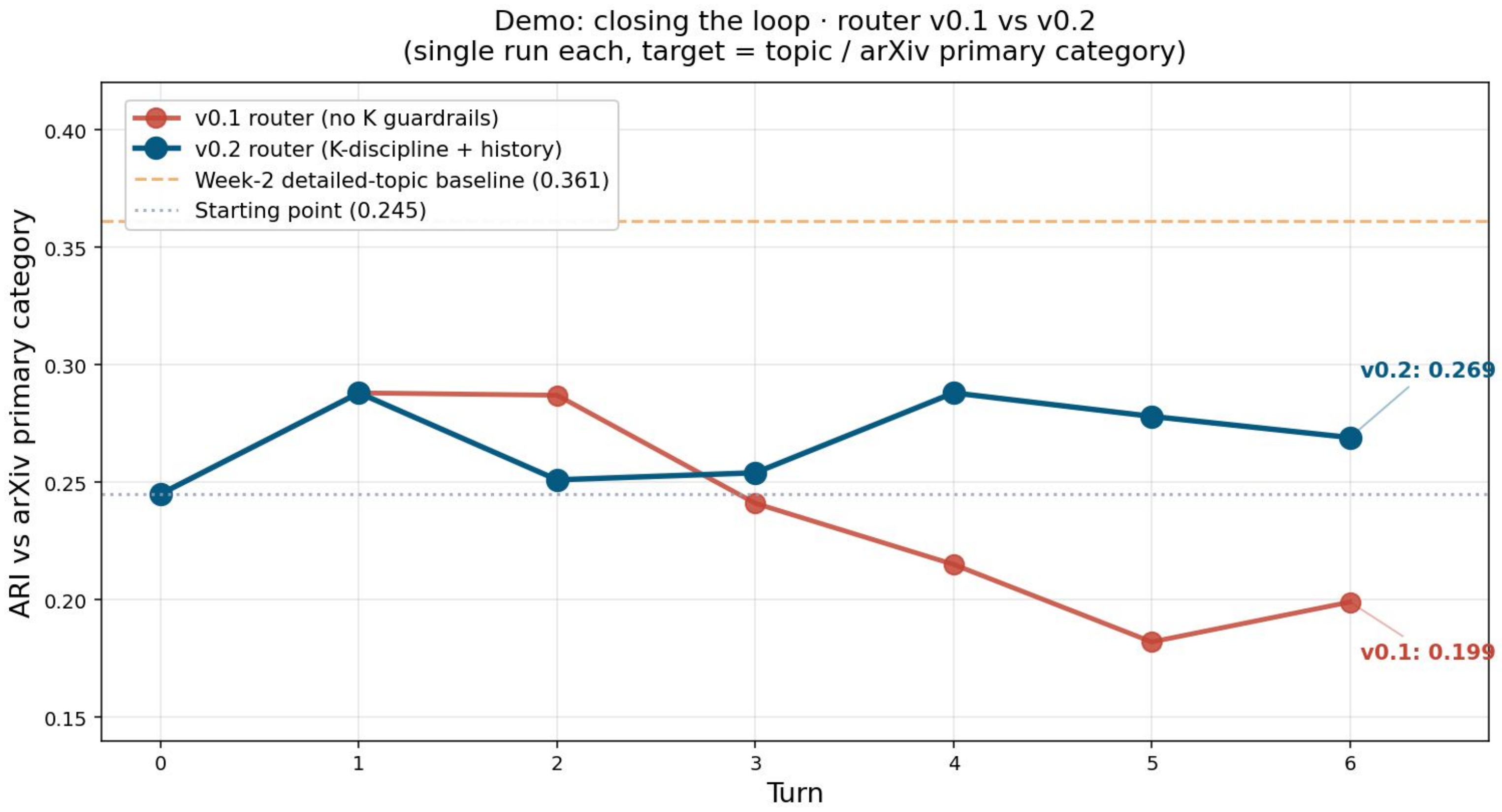
`move(items, c)`
manually reassign items

`re-embed(axis)`
re-encode emphasizing an axis

`ignore(pattern)`
strip terms before embedding

Demo · closing the conversational loop

Two iterations of the router · what works, what's next



V0.1 LESSON

Over-fragmentation

Router increased K at every negative feedback, with no awareness of past decisions. K went 5 ⟶ 22.

V0.2 FIX

K-discipline + history

Prompt guardrails on K, router sees its own past decisions, ops capped at 2/turn.

NEXT BOTTLENECK

Semantic split

split() partitions by k-means geometry, not by the criterion the user asks for.

Difficulties

TIME TRADE-OFF

Local-first cost

llama3.2:3b validated the pipeline but absorbed a sprint before cloud results came in. ARI ≈ 0.03 confirmed the model was insufficient — the right finding, but expensive in time.

ARCHITECTURAL TENSION

Semantic vs. geometric split

Cluster-level ops prevent the user from cheating with paper IDs. But the same design blocks semantically-targeted splits — k-means partitions by geometry, not by the criterion string.

FAILURE

LLM-LLM coordination

v0.1 demo K-exploded (5⟶22) and ARI dropped. Neither agent was individually wrong — the failure was in their interaction.

METHODOLOGICAL CHOICE

Targets without defined labels

Methodology/scale targets have no free labels. Hand-labeling is ~3h each; LLM-labeling needs an audit. (LLM-label + 30-sample audit for example)

Thank you